

SIDDHANT MOHANTY

Software Engineer | Backend Development | Java, C++, Spring Boot, SQL, AI Integration

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SUMMARY

Software Engineer with hands-on experience building scalable backend systems using Java, Spring Boot, REST APIs, PostgreSQL, and microservice architectures. Strong foundation in data structures, algorithms, object-oriented design, testing, and software engineering principles. Experienced in delivering production-grade features through Agile development practices.

EDUCATION

Malaviya National Institute of Technology, Jaipur

Nov 2022 – May 2026

Bachelor of Technology, Electronics and Communication Engineering | CGPA: 8.70/10

WORK EXPERIENCE

Arcesium India Pvt. Ltd.

May 2025 – Jul 2025

Software Developer Intern | Corporate Card Invoice Automation

- Architected and optimised **RESTful** backend microservices in **Java, Kotlin, and Spring Boot**, implementing modular service layers, and ACID-compliant database transaction workflows to fully automate corporate credit card invoice creation and payment processing.
- Engineered a comprehensive test suite using **JUnit 5 and Mockito**, achieving a **98%** QA pass rate across unit and integration test coverage; eliminated flaky test failures through mock isolation strategies.
- Eliminated **95%** of manually processed bottlenecks, delivering **\$30,000+** in annual operational cost savings and reducing processing time by **99.8%**.
- Collaborated cross-functionally with product and QA teams in an **Agile/Scrum** environment to scope, implement, and release features within tight sprint cycles.

PROJECTS

AI-Powered E-Commerce Platform

Tech Stack: Java, Spring Boot, Docker Compose, React, OpenAI, pgvector

- Designed, developed, and containerised a full-stack e-commerce application with **Spring Boot, PostgreSQL, Docker Compose**, implementing **RESTful CRUD APIs** for product catalogue management, shopping cart, and multi-step checkout flows, ensuring environment consistency across development and deployment.
- Integrated **OpenAI API** to automate AI-generated product descriptions and images, reducing manual content creation overhead for sellers by enabling single-click listing generation.
- Implemented a context-aware AI chatbot using **RAG** to answer real-time product and order queries, reducing support load through automated resolution.

Multimodal Conversational AI with Memory

Tech Stack: Java, Spring AI, OpenAI API, Docker Compose

- Engineered a multimodal AI chat backend using Spring Boot and the OpenAI API, supporting text conversation, image understanding, image generation, speech-to-text transcription, and audio processing in a unified pipeline.
- Integrated semantic search with **pgvector** to enable context-sensitive memory retrieval and improve conversational relevance.

JobApp Portal

Tech Stack: Spring Boot, Spring Security, PostgreSQL, React, Spring AOP, REST APIs

- Built a full-stack job application portal with role-based access control using **Spring Security**, supporting separate employer and applicant workflows with secure **JWT**-based authentication.
- Implemented request logging and performance monitoring using **Spring AOP**, while optimising PostgreSQL relational data models and indexed queries for efficient applicant filtering and structured audit trails.

SKILLS

Languages: C++, Java (OOP, AOP), Kotlin, JavaScript, SQL (PostgreSQL, MySQL, MS SQL Server), Python,

Frameworks & Web: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Spring AI, React, Hibernate, MyBatis

AI & ML: OpenAI API, Retrieval-Augmented Generation (RAG), Embedding Models, pgvector, Ollama

DevOps & Cloud: Docker, Docker Compose, Maven, AWS (EC2, S3), GitHub, GitLab, CI/CD, DBeaver

Testing: JUnit 5, Mockito, MockK, Unit Testing, Integration Testing, TDD

Coursework: Data Structures & Algorithms, System Design, OOP, Microservices, REST API Design, TCP/IP, HTTP, Operating Systems, Computer Networks, Agile/Scrum, SDLC.

LEADERSHIP EXPERIENCE

President, Cultural Society, MNIT Jaipur

Oct 2024 – May 2025

Led a 30+ member student council, overseeing end-to-end planning and execution of 15+ institute-wide cultural events, managing multi-lakh event budgets and cross-team coordination.

Student Training Coordinator, Placement & Training Cell, MNIT Jaipur

Jun 2024 – May 2025

Organised and facilitated 10+ placement preparation workshops and alumni-led training sessions, directly benefiting 200+ students with technical interview and career readiness skills.

CODING PROFILE & ACHIEVEMENTS

Codeforces: 1284 (Pupil) **LeetCode:** 1876 (Knight Badge, Top 5%)

Solved **1000+** DSA problems across CodeForces, LeetCode, GFG.

CaseQuesta (BITS Pilani, **Top 8/200+** teams) | ProductoPedia (SRCC, **Top 5/50+** teams)